



CAN LMICs
AFFORD THE
CLIMATE
TRANSITION?

UNDERSTANDING FINANCING GAPS AT THE COUNTRY LEVEL

AUTHORS: Hannah Luscombe ¹, Vivien Foster ², Will Collier²,
Miles Greene², Sam Fankhauser¹, Jairo Quirós-Tortós ³

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Abstract

Achieving the goals outlined in the Paris Agreement necessitates an increase in clean energy investments by the mid-century. Consequently, many low- and middle-income countries (LMICs) are poised to encounter considerable climate financing gaps, stemming from insufficient funds to repay the financing necessary for enacting net zero. This challenge is compounded by a lack of concessional resources, escalating debt obligations, and amplified borrowing expenses. A significant research gap regarding individual countries' financial capacities to meet repayment obligations linked with the net zero transition exists.

This study presents a novel analytical model, MinFin, that identifies energy sector financing gaps by projecting investment needs, estimating the associated cost of finance, and comparing this to anticipated sectoral cashflows. Additionally, the model facilitates the exploration of strategies to bridge climate financing

shortfalls through interactive simulations of alternative scenarios. Moreover, it is tailored to foster effective collaboration between energy and finance ministries to ensure that national energy planning processes comprehensively address macroeconomic and financial considerations. Traditionally, energy plans have been formulated without involvement from finance ministries.

The paper further demonstrates the feasibility and utility of the model through illustrative applications to Kenya and Morocco. Results highlight the likely severity of climate financing gaps in LMICs, and the associated need for measures that reduce the financing costs countries face, while also boosting the generation of sector revenues to fund repayment. All this points to the importance of integrating long-term energy planning efforts with the development of national climate financing strategies, necessitating closer collaboration between Ministries of Energy and Finance.

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1 INTRODUCTION

Extensive public discussion of ‘climate financing gaps’ tends to focus on the huge gap between global investment needs and aggregate capital flows for the energy transition. While such figures play an important role in global climate advocacy, they do not shed light on the more important question of whether individual countries are in a position to finance their respective climate investment plans.

This paper provides an alternative country-level perspective on the ‘climate financing gap’. According to this view, a country faces a ‘climate financing gap’ when it lacks the cashflow to cover the financing costs associated with the investment programme it needs to deliver the energy transition. This comes from the basic insight that there can be no financing without funding, as finance is only ever provided when there is proven capacity to repay.

The extent to which individual countries may face such a climate financing gap depends on the interaction between three factors: investment needs, financing requirements and funding availability. Investment needs convey how much of an increase in capital expenditure would be needed to meet climate goals – **the greater the carbon intensity of the current energy system, and the faster the growth in energy demand, the larger the investment required to meet net zero goals**. Financing requirements capture the terms on which the country can secure climate finance: countries with relatively elevated country risk premiums and/or relatively limited access to concessional financing will need to pay much higher financing costs to implement any given investment plan. Finally, funding availability refers to the resources available in the

sector to repay financiers: the more limited the public resource envelope, and the weaker the cashflow of public and private energy utilities, the more difficult it will be to repay financiers.

Beyond providing a useful conceptual framework, this approach can be relatively easily quantified to evaluate the extent to which specific countries face climate finance gaps, identifying where and when these gaps are likely to arise, and exploring measures that could be taken to reduce them. MinFin is an emerging model that takes any given investment plan for energy transition and computes the associated future annual financing requirements, based on the specific financing terms available to any given country. The model then compares annual financing requirements against that country’s projected funding availability, to determine whether implementation is financially feasible.

This approach can be particularly valuable to facilitate dialogue between Ministries of Energy, typically responsible for producing optimal climate investment plans based on engineering-economic paradigms, and Ministries of Finance, which carry responsibility for raising the associated climate finance (The Coalition of Finance Ministers for Climate Action, 2023). In particular, the proposed methodology supports a holistic examination of financing needs relative to available funding envelopes at the country level, allowing Ministries of Finance to take a strategic overview of the climate financing problem and to explore the full range of financing options in relation to the full portfolio of clean energy projects (Luscombe et al., 2024).

The remainder of this paper is organised as follows. Section 2 situates this work in the wider literature. Section 3 sketches out the conceptual framework underpinning the MinFin model that permits the quantification of climate finance gaps at the country level. Section 4 discusses

the methodological challenges of empirically calibrating the model. Section 5 presents the results of two illustrative and contrasting studies that calibrate the model for Kenya and Morocco, capturing a range of low- and middle-income country settings in Africa.

2 LITERATURE REVIEW

There is a pressing need to significantly boost global investment for mitigating and adapting to climate change. Much of this investment will be frontloaded and require long-term, capital-intensive commitments, particularly for sustainable infrastructure in energy, buildings, and transportation. A large literature exists regarding the global investment requirements to constrain global warming; Stern et al. (2021) project US\$4 trillion annually by 2030 for Paris Agreement goals and the International Energy Agency (IEA, 2021) envisions US\$5 trillion annual global energy sector investment by 2030. CPI (2023) identifies a gradual rise of up to US\$10 trillion annually by 2050. In low- and middle-income countries (LMICs), estimates from the IEA (2023a) and Songwe, Stern, and Bhattacharya (2022) indicate a need for around US\$2.8 trillion per year in clean energy investment by the early 2030s.

However, the existing investment landscape falls short, providing just USD 1.3 trillion of climate finance in 2021. Furthermore, capital flows exhibit geographical concentration, with developed economies leading in the mobilisation of funds (CPI, 2023). Moreover, despite recent growth in climate finance, current flows still make up no more than 1 percent of global GDP. Typically, discussions surrounding financing gaps centre on this disparity between global investment

requirements for the energy transition and reported aggregate capital flows.

Such estimates fail to account for two critical factors. First, the cost of capital associated with the implementation of climate investment plans will vary significantly according to the specifics of local markets and technologies. Second, countries will not necessarily have the cashflows needed to support the repayment of such finance.

Recent empirical research finds substantial variations in the cost of capital between countries (Angelopoulos et al., 2016; Egli, Steffen, and Schmidt, 2019), with developing countries exhibiting a higher average cost of capital compared to industrialised nations. Even within countries, structural differences between different renewable energy technologies, as well as changes in the cost of capital over time due to technology-specific effects or economy-wide interest rate fluctuations, are evident (Egli, Steffen, and Schmidt, 2018; Kirkpatrick and Benneer, 2014). Steffen (2020) analysed the cost of capital for solar PV, onshore wind, and offshore wind across 46 countries, finding that a consistent rank order among technologies emerged. This highlights the potential underestimation of financing requirements, particularly in regions with high costs of capital, such as LMICs. Furthermore, Agutu et al. (2022) find that when energy planning models account for country- and

mode-specific changes in the cost of capital, the optimal choice of electrification modes changes substantially. These research findings support analysis in the policy literature suggesting that a country with high financing costs can witness a 40 percent increase in wind investment costs relative to international benchmarks, while under the same conditions, gas may only become 6 percent more expensive (UNDP, 2013).

At the same time, the existing literature highlights a notable gap in research concerning the financial capacities of individual countries to fulfil repayment obligations associated with the energy transition (Polzin, Sanders, and Serebriakova, 2021). Polzin and Sanders (2020) reveal a qualitative mismatch between the demand for financial resources and the potential supply, emphasising the need for a more nuanced understanding of financial dynamics in achieving energy transition goals. Building on this, Polzin, Sanders, and Serebriakova (2021) introduce a methodology that translates projected investments in global transition scenarios into identifiable sources of finance. However, the exploration of the optimal mix of financing needs for energy transition scenarios remains largely unexplored, as highlighted by McCollum et al. (2018).

When developing and executing long-term strategies, emerging literature emphasises the importance of fostering close collaboration between Ministries of Energy and Ministries of Finance (Coalition of Finance Ministers for Climate Action in 2023). Ministries of Finance play a crucial role in coordinating sector ministries and other state agencies to implement strategies (Elliott et al., 2019).

These ministries should actively participate in national planning processes to ensure that plans address macroeconomic and finance-related considerations. It is essential to assess costs adequately, consider broader financial implications, and integrate plans into overall budget management processes (The Coalition of Finance Ministers for Climate Action, 2023).

The experience of pioneering countries, such as Costa Rica, has highlighted the importance of a robust and iterative modelling approach and the inclusion of cross-sectoral stakeholders as key elements in a process that allowed the Ministry of Finance to mobilise billions of dollars of concessionary resources for the climate transition progressing 'Data-to-Deal' (Jaramillo et al., 2023; Luscombe et al., 2024; Jaramillo and Saavedra, 2021). This Data-to-Deal approach calls for cross-ministerial dialogues designed to eliminate silos within government ministries and improve the integration between energy planning and associated financing strategies.

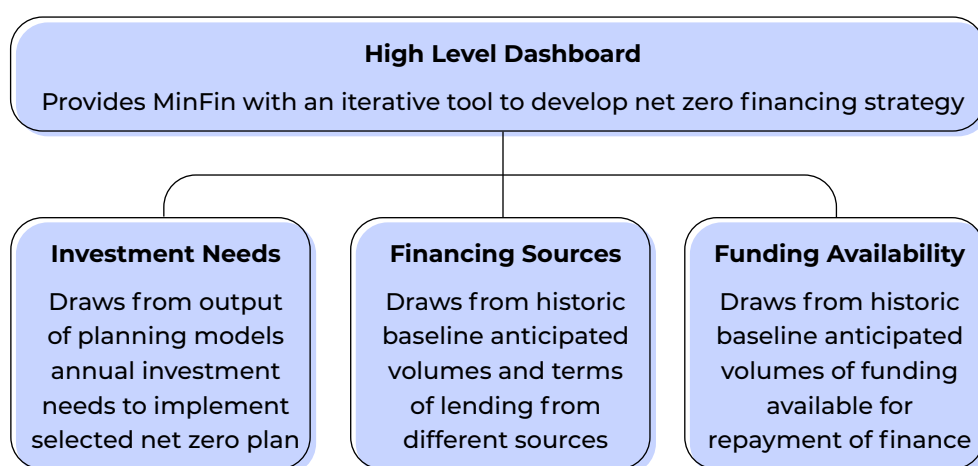
Modelling tools play a valuable role in this process, particularly when these are open-source, heightening transparency. Tan et al. (2024) propose a methodology for an innovative whole-systems energy modelling approach that integrates open-source models in an analytical workflow with the ultimate aim of mobilising financing and developing financially viable investment plans. However, despite the existence of numerous analytical models that support different aspects of the energy transition, there is currently no model specifically designed to facilitate dialogue among Ministries of Finance for evaluating the overall financial viability of long-term energy plans.

3 CONCEPTUAL FRAMEWORK

MinFin is a simple model that can be used to support the development of national level climate finance strategies, building on three

pillars that come together in a dashboard providing a simple visualisation of a country's projected climate finance position (Figure 1).

Figure 1: Conceptual structure of the MinFin model.



The first pillar is investment needs. Given that clean energy technologies are significantly more capital intensive than fossil fuel alternatives (Best, 2017), reaching net zero goals by mid-century will necessitate a surge of investment relative to historic levels. System planning models, such as OSeMOSYS, provide annual investment needs to meet specified energy transition goals, such as mid-century net zero (Howells et al., 2011).

However, classical engineering-economic planning models, like OSeMOSYS, are likely to underestimate the financing needs for the energy transition, because they only estimate the cost of building new plants. Although such models also stipulate premature withdrawal of fossil fuel plants, when this is called for to meet

climate targets, these shutdowns are treated as costless, because the plants have already been built. In practice, early shutdown of fossil fuel plants is only possible if owners are financially compensated for foregone profits over the remaining life of the assets or associated power purchase agreements. Tools such as the World Bank's Fossil Fuel Retirement Model can be used to estimate the magnitude of the requisite compensation based on several basic input parameters, such as the age and cost structure of the plant and the associated contracting arrangements (Suski et al., 2022).

The second pillar is financing sources. Finance provides a critical means of spreading out the investment peak over a longer time-period.

Countries may be able to access a variety of sources of finance, ranging from domestic to international and commercial to concessional, each with widely varying financial terms. Such terms are defined by the rate of interest, the maturity of the loan, and the presence of any grace period before repayment begins. A limited amount of concessional finance, available from international financial institutions, typically softens financing terms through a combination of lower interest rates, longer maturities, and larger grace periods. The discount that this represents relative to market benchmarks is known as the grant element.

Terms of finance will also differ significantly across countries and technologies, reflecting the particular risk profiles presented in each case. Based on an exhaustive compilation of the terms at which a country has been able to access finance for clean energy projects in the last decade, it is possible to estimate the typical weighted average terms of available finance. These can be used to project forward the financing payments associated with the implementation of the energy transition plan.

The third pillar is funding sources. There can be no financing without funding, as finance is only ever provided when there is proven capacity to repay. In that sense, it becomes important to determine what resources a country may have available to repay clean energy finance. Typically, the main sources of cashflow available for the repayment of investments in clean energy are public investment and free cashflow from energy utilities. The allocation of public investment to the energy transition is primarily a political choice, reflecting the nature of competing spending priorities in each country. The available cashflow from energy utilities will largely reflect the adequacy of end-user tariffs, as well as the operational and financial efficiency of the enterprise. In addition, for countries that are

heavily dependent on fossil fuels, the gradual substitution of fossil fuel technologies with clean energy solutions may generate significant savings in fuel expenditure, which could potentially be redirected towards repayment of financing for clean energy development. However, the feasibility of doing so may depend on whether the institutions accruing the fossil fuel expenditure savings are the same as those liable for repayment of clean energy projects, and – if not – whether it is possible to design transfers between them. In addition, lower income countries may be able to benefit from some international grants that can be considered a further addition to the funding envelope. The projection of available funding envelopes into the future is particularly challenging but can be done either based on extrapolation of historic funding trends or by holding constant the share of GDP available for sector funding.

Lastly, the high-level dashboard brings together these three pillars into a simple visualisation of future climate financing gaps. To estimate these, future financing requirements are first estimated, by applying historic financing terms to the future net zero investment plan (see black line in Figure 2). This determines the annual repayment that would be needed to service such a debt profile under business-as-usual conditions (see red line in Figure 2). Next, funding availability is also projected forward based on the extrapolation of historic funding trends (see green line in Figure 2). By comparing these two projections against each other, the presence of a climate financing gap can be established, whenever the projected financing requirement (represented by the red line) exceeds the funding availability (represented by the green line).

In the MinFin dashboard below, the left-hand panel illustrates a business-as-usual situation in which the country continues to access financing

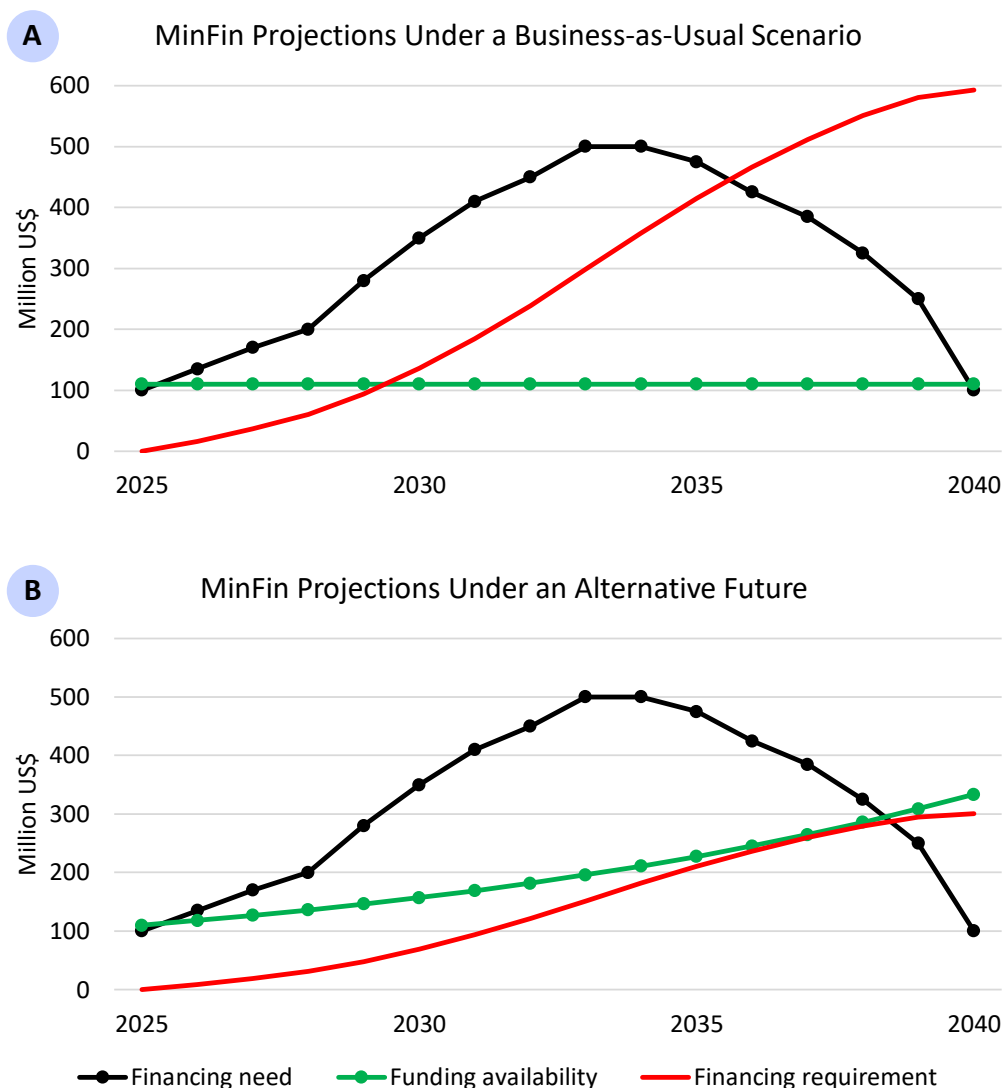
on the same terms as historically, while funding streams are extrapolated to remain constant based on recent trends. Under these assumptions, financing requirements climb exponentially (the red line in Figure 2a), rapidly breaching the available funding envelope (the green line in Figure 2a) by 2030.

The right-hand panel illustrates how the MinFin model can be used to simulate alternative futures that would be needed to close country climate financing gaps. In this example, it is supposed that the country can substantially improve on its weighted average terms of finance, perhaps by accessing more concessional resources and/or taking measures to reduce country and/or sector risk. This leads to a substantial flattening of the required financing repayments (the red line in Figure 2b), even though the underlying investment plan remains unchanged (the black line in Figure 2b). At the same time, it is assumed that the country is able to substantially boost the different sources of funding available to the sector, raising the compound annual growth rates (CAGR) from zero to 5–10%, leading to a major shift of the funding envelope (the green line in Figure 2b). As a result, the climate financing plan remains affordable through to mid-century, with the funding envelope comfortably exceeding the required financing repayment.

This simple example illustrates that country-level climate financing gaps arise from a misalignment between the ambition of the climate investment plan, the softness or hardness of terms on which finance can be accessed, and the strength of available cashflows in the sector. At the same time, countries with relatively elevated country risk premiums and/or relatively limited access to concessional financing will need to pay much higher financing costs to implement any given investment plan. Likewise, the more limited the public resource envelope, and the weaker the financial position of energy utilities, the more difficult it will be to repay financiers. By the same token, climate finance gaps can be eased, and potentially eliminated, by adjusting one or more of these levers.

The analysis can be further enriched through the comparison between a net zero investment scenario and a conventional Least Cost scenario that continues to incorporate fossil fuels. This permits a cost benefit analysis where the incremental capital cost associated with net zero can be compared against the present value of reduced system operating costs, as well as externality benefits associated with lower emissions of carbon (and potentially other local pollutants). There is also the potential for future work to incorporate a fiscal module that computes the impact on tax receipts and subsidy expenditures resulting from shifts in the pattern of energy consumption under the transition to net zero.

Figure 2: MinFin dashboard contrasting climate financing gap under alternative projections. a) MinFin projections under a business-as-usual scenario, b) MinFin projections under an alternative future.



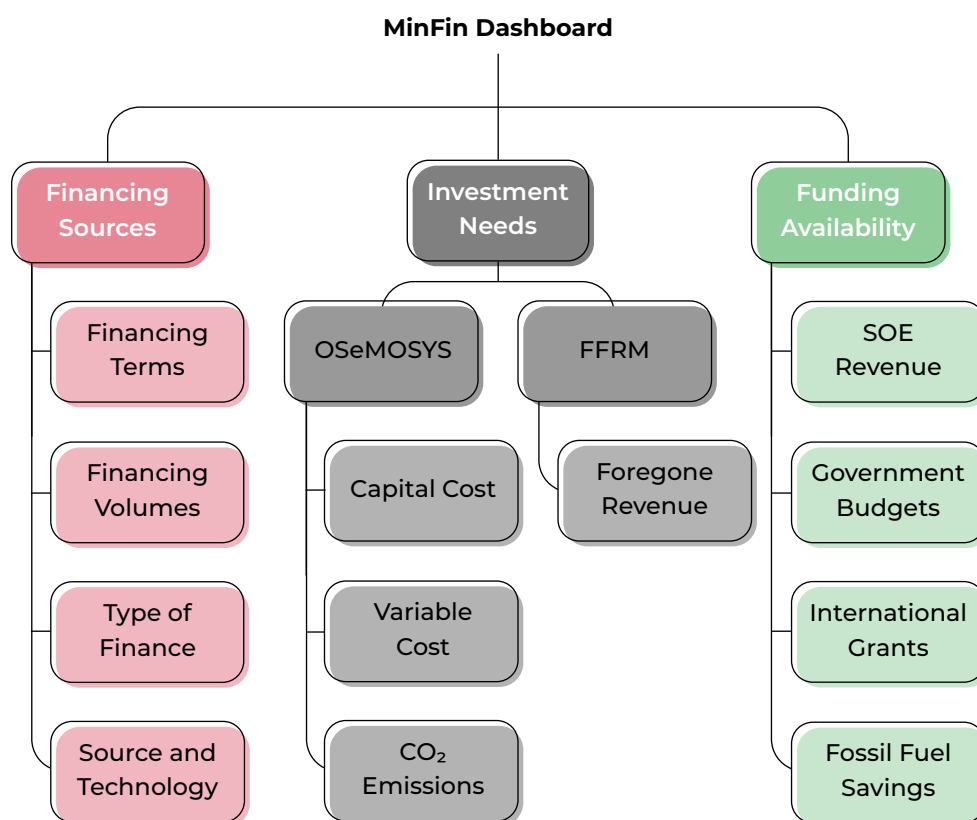
	Past	Future
Financing terms		
Interest rate	7%	4%
Term of loan	13	25
Funding CAGR		
Public spending	0%	5%
State-owned enterprise cashflow	0%	10%
Fossil fuel savings	0%	10%
International grants	0%	5%

4 EMPIRICAL CALIBRATION

In addition to being conceptually intuitive, the MinFin model is also relatively easy to calibrate from publicly available data sources. This section

describes the methodology involved in the empirical calibration of the model, which can be broken down into various building blocks as illustrated (Figure 3).

Figure 3: Building blocks associated with empirical calibration of the MinFin model. Note: OSeMOSYS = Open Source energy MOdelling SYStem; FFRM = Fossil Fuel Retirement Model; SOE = State-owned Enterprise.



Investment Needs

The calibration process begins by determining the financing requirements for achieving energy transition objectives, notably net zero emissions by 2050. To estimate financing requirements, MinFin draws inputs from two other energy models. First, an energy planning model (in this case OSeMOSYS) is used to identify the optimal

year-on-year investment plan needed to satisfy projected demand growth, broken down in considerable detail based on 23 different power technology types (Howells et al., 2011). Second, a simple financial model – in this case the World Bank’s Fossil Fuel Retirement Model (better known as the Coal Retirement Model) – is used to estimate

the payments required to compensate asset owners for the foregone profits arising from the premature withdrawal of fossil fuel plants (Suski et al., 2022). These two inputs are aggregated within MinFin to determine the total capital requirement for net zero transition year on year.

OSeMOSYS is a widely used open-source systems optimisation linear model for long-term energy system planning, which includes a detailed representation of the power sector nested within a more comprehensive representation of the wider energy system. OSeMOSYS represents energy systems via sets of technologies and energy carriers, with the objective of identifying the lowest net present value (NPV) of an energy system that will meet specified demands for energy (Howells et al., 2011). Outputs from OSeMOSYS provide important input parameters for MinFin, notably capital, variable, and fixed costs, year on year, at technology level granularity, as well as the trajectory for annual aggregate energy system carbon emissions.

For MinFin purposes, OSeMOSYS is used to generate optimal investment plans for two distinct scenarios. A baseline Least Cost (LC) scenario is first established, for which OSeMOSYS simply selects the least cost combination of projects needed to meet demand projections, without considering carbon constraints. Depending on the country, this scenario may include a significant fossil fuel component. For the other Net Zero (NZ) scenario, OSeMOSYS instead satisfies demand projections subject to the constraint of eliminating carbon emissions from the power system by 2050. While MinFin analysis focuses primarily on the NZ scenario (or any preferred energy transition scenario), the ability to compare and difference NZ and LC results is important, since it permits the estimation of incremental capital costs and associated

operating cost savings under net zero, as well as computation of carbon emissions saved and the associated implicit cost, which is of potential interest from a carbon markets perspective (Strand, 2023; Piris-Cabezas, Lubowski and Leslie, 2019; Edmonds et al., 2021).

A cost of carbon can be applied to CO₂ emission results derived from OSeMOSYS. The assumed carbon cost aligns with the 2017 High-Level Commission on Carbon Prices, utilising midpoints between its high and low estimates [Appendix A]. Since the High-Level Commission report fails to specify the social value of carbon beyond 2030, projections from 2030 to 2050 are extrapolated using the same consistent growth rate of 2.25 percent per year (Stiglitz et al., 2017).

The Fossil Fuel Retirement Model (FFRM) was originally developed by the World Bank to support the estimation of asset stranding costs for early retirement of coal plants, but can readily be adapted to perform the same process for other fossil fuel plants, based on oil or gas. The FFRM draws on OSeMOSYS outputs to identify all fossil fuel plants that need to be shut down in the NZ scenario prior to reaching the end of their economic life. The FFRM calculates the net revenue foregone by the investor due to early shutdown, using OSeMOSYS data on plant commissioning year, energy generation, plant load factor, economic life of plant, as well as capital and operating costs. In addition, FFRM projects plant revenues based either on Power Purchase Agreements (PPAs), where these are in place with known contractual terms, or market price, where plants operate freely in the context of a wholesale market. In settings where neither of these assumptions apply, the sale price of energy can be approximated by the system marginal cost calculated in OSeMOSYS. For MinFin purposes, the present value of the foregone net revenues of stranded fossil fuel plants is added to the country's net zero financing requirement.

Financing Sources

The construction of a financing baseline entails the compilation of a database containing detailed information on financing terms provided by specific financiers for the full set of energy projects implemented in a country over the last decade [Appendix B]. The purpose is to understand, from the historical evidence, what has been a country's track record in attracting financing from different sources and on what terms.

The process begins with the creation of an inventory of all energy projects commissioned during the baseline period, based on a reputable public source, such as the WRI Global Power Plant Inventory (WRI, 2018). Each of these power plants is then extensively researched to ascertain the total investment cost, the different entities that contributed to its finance, and the terms on which finance was provided. Much relevant information is already available in global public databases of energy financing flows, including IEA's World Energy Investment Database (IEA, 2023b), The International Renewable Energy Agency's Renewable Energy Public Investment Database (IRENA, 2023), the CPI Global Climate Finance Landscape (CPI, 2023), the World Bank's Private Participation in Infrastructure Database (World Bank, 2023a), and Boston University's China Africa Loan Database (Boston University, 2023). In addition, targeted internet research is used to locate relevant documents for each project and address data gaps. Where project-specific data cannot be found, gaps can be filled either by using averages for equivalent project categories within the same database, or by lifting relevant benchmarks from the literature.

The final output is a comprehensive database that itemises each financier's share in each project and furnishes details regarding the financing terms. These terms encompass elements like interest rates, grace periods, loan maturities, loan currency, and the loan repayment schedule.

While debt financing holds a significant role, it is equally important to record all pertinent equity contributions. This database provides an informative descriptive overview of a country's historic energy financing patterns. It is also used to generate useful summary statistics regarding overall financing volumes and terms for energy projects in the country, as well as the disaggregated financing volumes and terms by different project or financier characteristics. Indicators used to capture the hardness or softness of financing terms are the Weighted Average Cost of Capital (WACC) and the market element. Following Scott (2017), the market element is defined as the ratio of the present value of the repayments under a loan repayment schedule under current terms divided by the present value of the repayments for an equivalent loan offered under market terms; equivalent to 100 minus the grant element [Appendix C].

Once the financing baseline is established, MinFin uses the overall historic WACC to project the future annual financing requirement for the obligations arising from OSeMOSYS and FFRM. Each year, as new investment projects and stranded plant compensation liabilities arise, MinFin assumes that these will be financed on historic terms and projects each of them into a stream of year-on-year future financing repayments, based on standard financial formulas. These are added to continuing financing repayments from ongoing projects contracted in preceding years. In this way, the country's established financing terms are used to simulate how future obligations will be spread over time. Given that the future will not necessarily match the past, this represents a baseline projection that can be subsequently modified and further explored.

Funding Availability

The construction of a funding baseline entails the compilation of data from multiple sources, to capture the resource envelope that has historically been available to a country to fund repayment

of finance. In practice, this mainly comprises government budgets and state-owned enterprise (SOE) cashflow. In some cases, international grants may also be relevant.

Public investment in the electricity sector is typically accessible through official Government Budget Reports, with supplementary data found in global public databases like the World Bank's comprehensive BOOST database (World Bank, 2023b). Care must be taken to identify the relevant disaggregated budget lines that pertain to direct investments in energy infrastructure and/or capital subsidies to SOEs. A further potential complication arises in countries where Appropriations in Aid are integrated into the public budget, raising the risk of double counting as aid projects are already captured under the financing baseline.

Cashflow statements for energy utilities (whether public or private) are typically published and allow extraction of data on free cashflow from operations, which indicates the potential budget available to finance investment. The World Bank's UPBEAT database, which compiles power utility financial statements for numerous developing countries, provides a data source. Depending on the power sector structure in the country, data on free cashflow from operations may need to be aggregated across several enterprises. For countries where financial reporting is deficient, alternative proxy figures may be available from

annual reports, such as historic volumes of capital expenditure.

Once the funding baseline is established, MinFin projects this forward as an estimate of the envelope of resources available to repay future financial obligations in the sector. There is limited hard evidence on which to base such projections. A reasonable starting point may be to simply extrapolate historic trends in funding, or to maintain the funding share of GDP and roll forward based on GDP growth projections published by the International Monetary Fund (IMF). In either case, the time horizon of net zero analysis will typically far exceed that of reputable economic forecasts. However, given that the future will not necessarily be a continuation of past trends, this simply represents a baseline projection that can be subsequently modified and further explored.

An additional consideration for countries transitioning away from heavy fossil fuel dependency is that fuel savings arising from the NZ Scenario may become substantial over time. These potentially represent another source of funding that could be captured and redirected towards repayment of clean energy finance, although this would depend on the financial and institutional arrangements in place in a country. The magnitude of such savings can be readily estimated through OSeMOSYS, by comparing fossil fuel expenditures between LC and NZ Scenarios, which can then be added to the projected funding envelope.

5 CASE APPLICATIONS

This section describes illustrative applications of the MinFin model in Kenya and Morocco, based on an extensive country level data collection process. Not only do these serve as a proof of concept for

the feasibility of implementing this approach, but they additionally provide emerging insights into the scale of the climate finance challenge and how it differs across low- and middle-income countries.

Kenya

Kenya has made strides in decarbonising its energy sector with close to 90 percent of its power capacity coming from abundant renewable sources (Brown, 2022; Oludhe, 2008; Oloo et al., 2015). However, given the anticipated growth in population, economy, and energy demand, there are substantial ongoing needs for investment in clean energy infrastructure to keep pace (Akrofi, 2021).

The OSeMOSYS model, adopted by Kihara et al., (2023), was employed to forecast investment requirements for Kenya. Two scenarios were simulated.

The LC Scenario aligns with the least-cost Power Development Plan (LCPDP) reference projections. Residual capacity, committed, and candidate power plants adhere to the plans outlined in the LCPDP. Additional potential from renewable sources becomes available after 2025, with a progressive exponential growth in capacity, while nuclear power does not become an option until 2036 (Kihara et al., 2023).

The NZ Scenario involves a gradual phase-out of natural gas capacity from 2045 to 2050 as well as a constraint of zero emissions by 2050. Up to the

mid-2030s, geothermal energy plays a pivotal role in meeting the burgeoning demand. Thereafter, wind deployment increases significantly, driven by growing cost competitiveness, and necessitating the integration of storage solutions to enhance system flexibility.

OSeMOSYS NZ results are used to determine the future investment needs for Kenya. These soar from a baseline level of US\$0.6 billion in 2025 to exceed US\$4 billion by 2050 and continue to grow beyond the net zero deadline reaching nearly US\$6 billion in the 2060s (Figure 4). Since fossil fuels no longer play a significant role in Kenya's energy mix, compensation required for the early shutdown of plants is negligible in this case.

A comparison of the two scenarios illustrates that capital costs associated with NZ are comparable to LC until mid-century but become significantly larger by about US\$100–200 million per year thereafter (Figure 5). This is a relatively small difference in percentage terms and is moreover compensated by the resulting benefits in terms of substantial savings in operating costs from renewable energy technologies, but above all in terms of the value of the resulting carbon emission savings.

Figure 4: Technology-specific investment needs for achieving net zero in Kenya by 2050 as projected by OSeMOSYS.

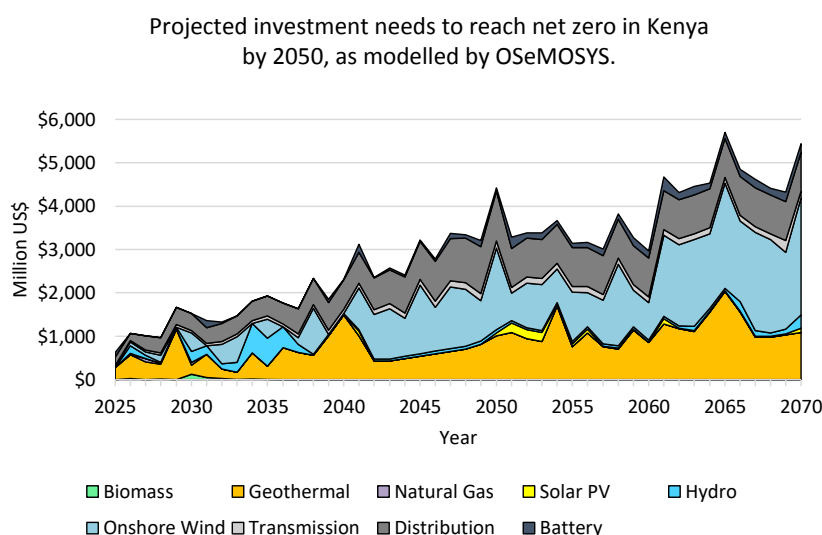
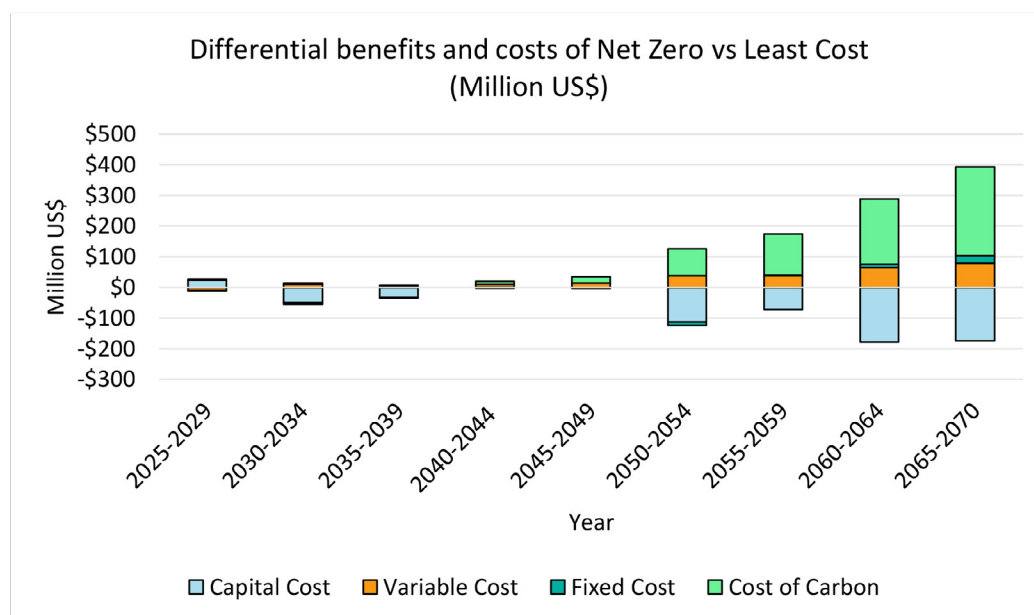


Figure 5: Differential benefits and costs of Net Zero vs Least Cost Scenarios in Kenya, 5-year average. This graph illustrates the cost difference between the Net Zero and Least Cost scenarios. Negative values represent additional costs associated with transitioning to net zero, while positive values indicate savings. When savings exceed costs, the transition to net zero is considered a favorable value proposition for the country.



A detailed review of historic terms of loan financing for energy projects in Kenya over the period 2010–2023 finds that the country has on average attracted US\$628 million of finance for clean energy annually. Clean energy finance in Kenya has overwhelmingly come from international concessional sources (US\$574 million), with softer financing terms, discounted at around 45–55 percent against market rates (Figure 6). By contrast, relatively little commercial finance has been going into Kenya’s energy sector, and that has been mainly from international (US\$35 million per year) rather than domestic (US\$4 million per year) sources.

The weighted average financing terms for both debt and equity in Kenya’s energy sector throughout the past decade featured an interest rate of 5.5 percent, coupled with a maturity period of 25.6 years and a grace period of 4.6 years. However, these terms exhibit notable variations across technologies. Overall non-renewable technologies have faced a higher cost of capital than their renewable counterparts, and across all technologies, the cost of equity (6–18 percent) substantially outstrips that of debt (2–10 percent) (Figure 7).

Figure 6: Merit order for historic finance sources of debt in Kenya, 2010 to 2023. Notes: Conc Dom = Domestic Public Sector concessional loans; Conc Intl = International Financial Institution concessional loans; Com Dom = Domestic commercial loans; Com Intl = International commercial loans.

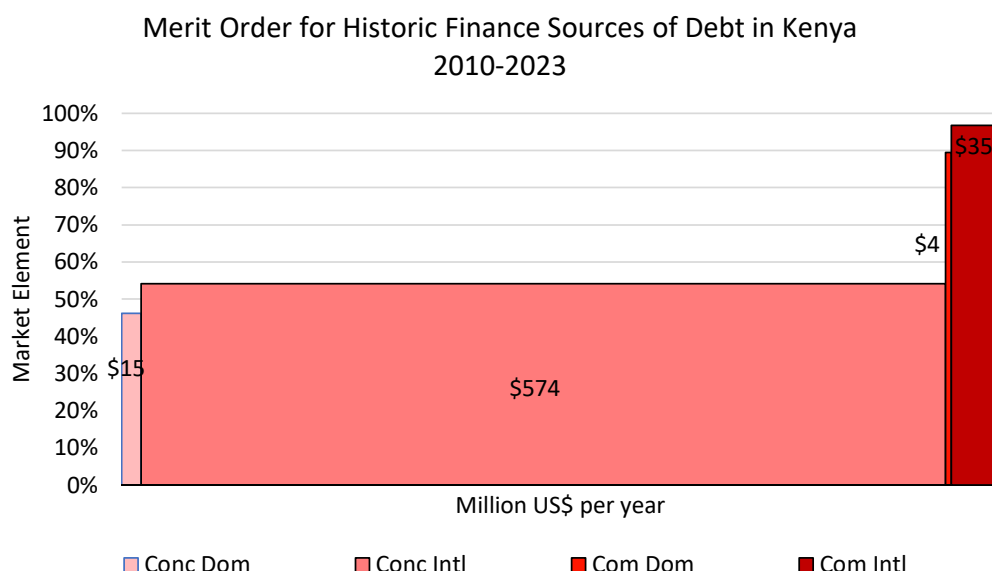
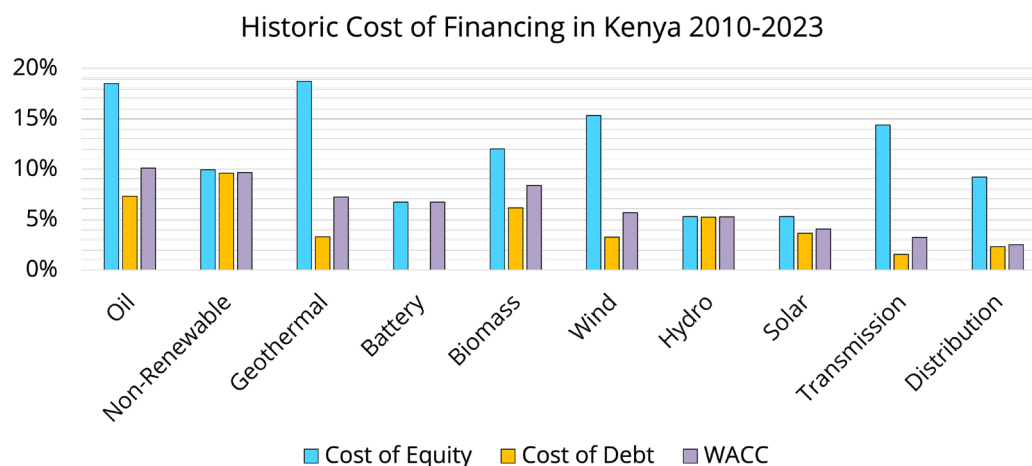


Figure 7: Historic Cost of Financing in Kenya based upon the period from 2010 to 2023. Notes: WACC= Weighted Average Cost of Capital.



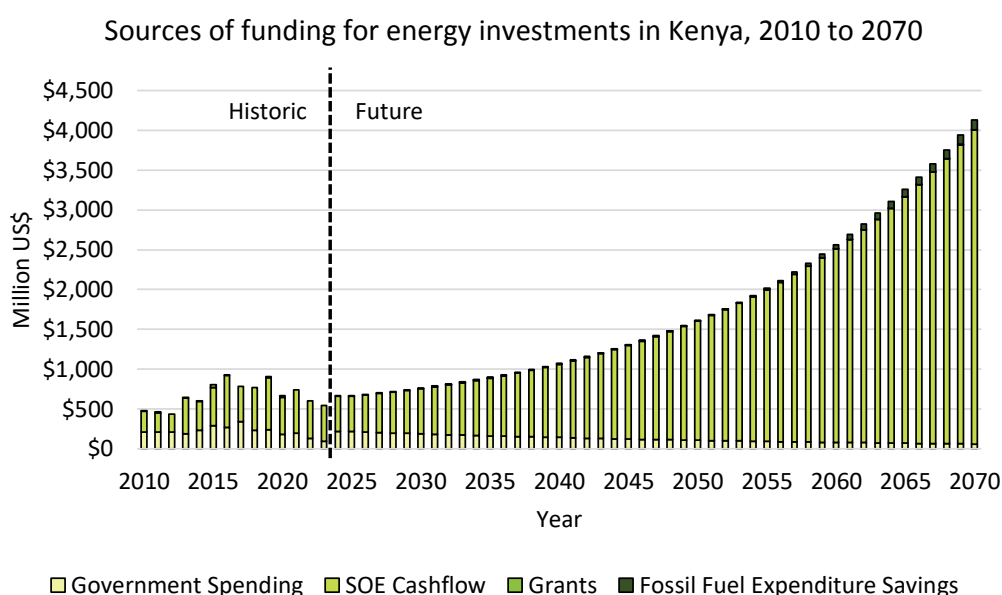
Analysis of historic cashflows in Kenya underscores the extent to which public budget allocations have been in decline (by close to 3 percent annually) and highlights the importance of cashflow from energy utilities (growing at 5 percent annually) as the main source of funding for the sector, with

international cash grants playing a negligible role (Figure 8). Overall, cashflows have been declining from a peak of some US\$900 million in 2016 to under US\$600 million by 2023, not even keeping pace with GDP growth over the same period. Should these trends persist, SOE internal

cash generation can be expected to be the primary source of funding for the sector (Figure 8). Due to Kenya's limited reliance on fossil fuels,

expenditure savings from the switch to renewable energy are modest in this case, albeit reaching US\$125 million per year by 2070.

Figure 8: Historic sources of funding and projected funding availability for energy investments in Kenya. Notes: SOE = state-owned enterprise.



Bringing together these elements into the MinFin dashboard for Kenya, highlights that a net zero transition cannot be financed under business-as-usual financing and funding conditions. This is because, beginning in 2025, Kenya starts to experience a gap between financing needs and funding flow, which grows continuously to reach US\$1.4 billion per year by 2050 and US\$2.4 billion per year by 2070 (Figure 9). These findings point to the need for improving upon historic financing conditions, and/or boosting sector cashflows, if the net zero transition is to be successfully implemented.

However, modifying solely the financing terms without increasing sector funding only postpones the climate finance gap. For example,

transition from the historical interest rate of 5.5 percent (with an average grace period of 4.6 years and a finance terms averaging 25.6 years) to an interest rate of 3.0 percent (with an average grace period of 5.5 years and a finance term averaging 30.0 years) would delay the gap until the 2040s (see Figure 10). At the same time, merely expanding funding sources to align with GDP growth projections of 5.5 percent annually only bridges the financing gap up until 2040, leaving a sufficient shortfall thereafter up until 2060 (see Figure 11). Simultaneously enhancing both historical financing conditions and bolstering sector cashflows makes the net zero transition viable, effectively closing the financing gap and rendering the transition economically feasible for Kenya (Figure 12).

Figure 9: MinFin projections in Kenya in absolute terms. Note: The weighted average interest rate is 5.5%, the average grace period is 4.6 years, and the average term of finance is 25.6 years. The CAGRs for government spending, SOE, and grants are -2.8%, 5.0%, and -6.5%, respectively.

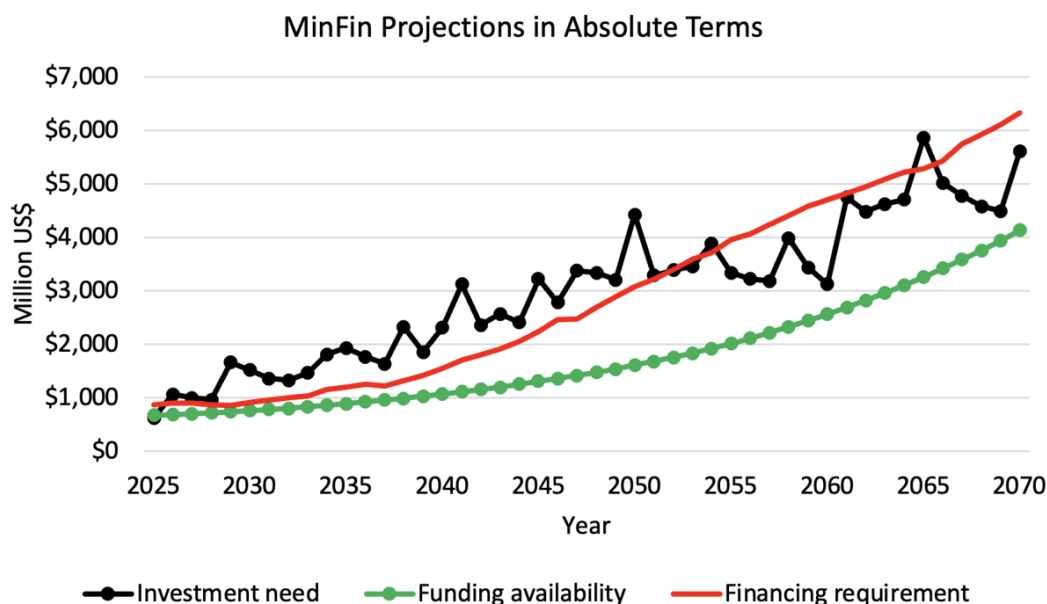


Figure 10: MinFin projections in Kenya with improved financing conditions. Note: The weighted average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 30.0 years. The CAGRs for government spending, SOE, and grants are -2.8%, 5.0%, and -6.5%, respectively.

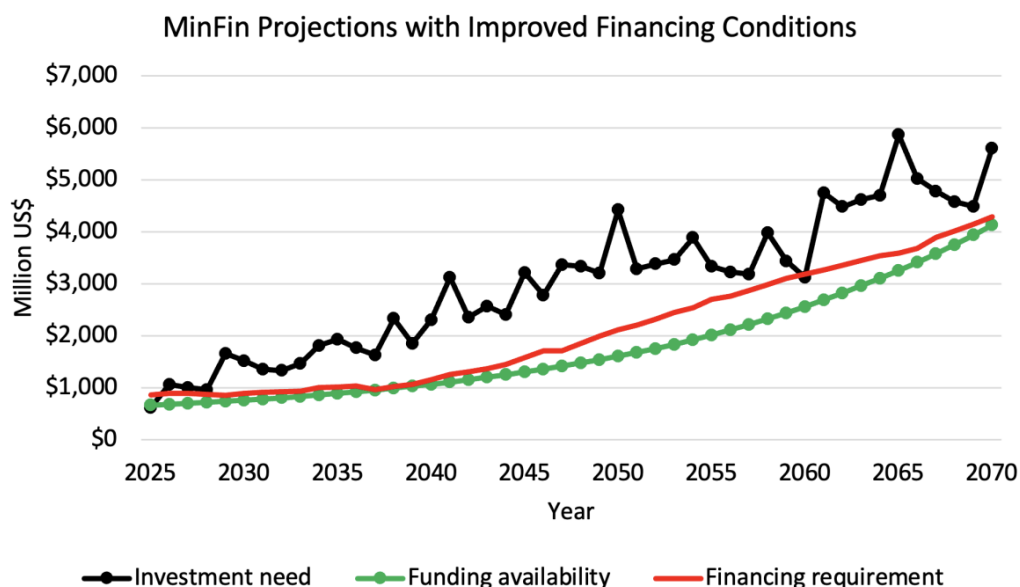


Figure 11: MinFin projections in Kenya with increased funding availability. Note: The weighted average interest rate is 5.5%, the average grace period is 4.6 years, and the average term of finance is 25.6 years. The CAGRs for government spending, SOE, and grants are 5.5%, 5.5%, and 5.5%, respectively.

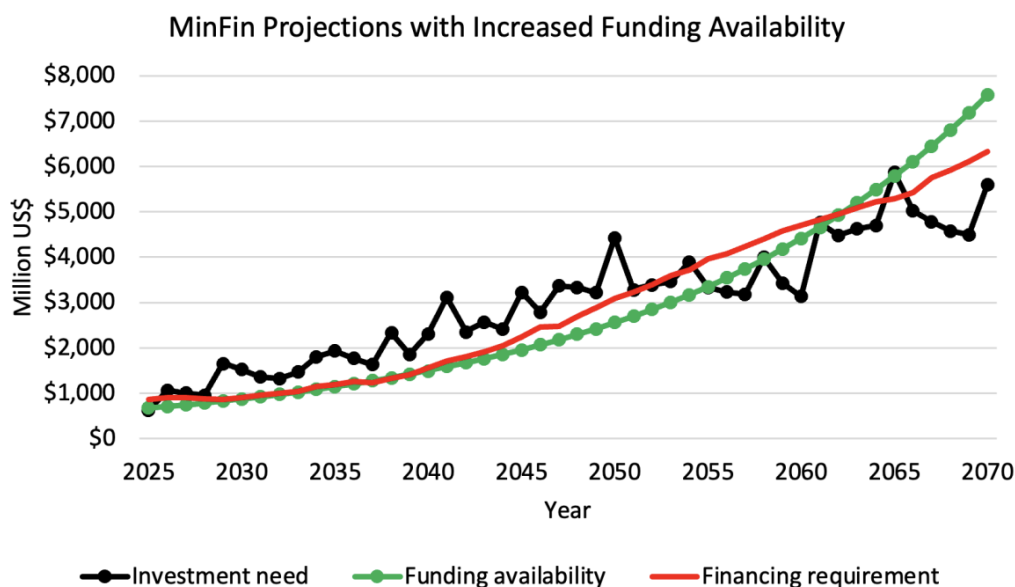
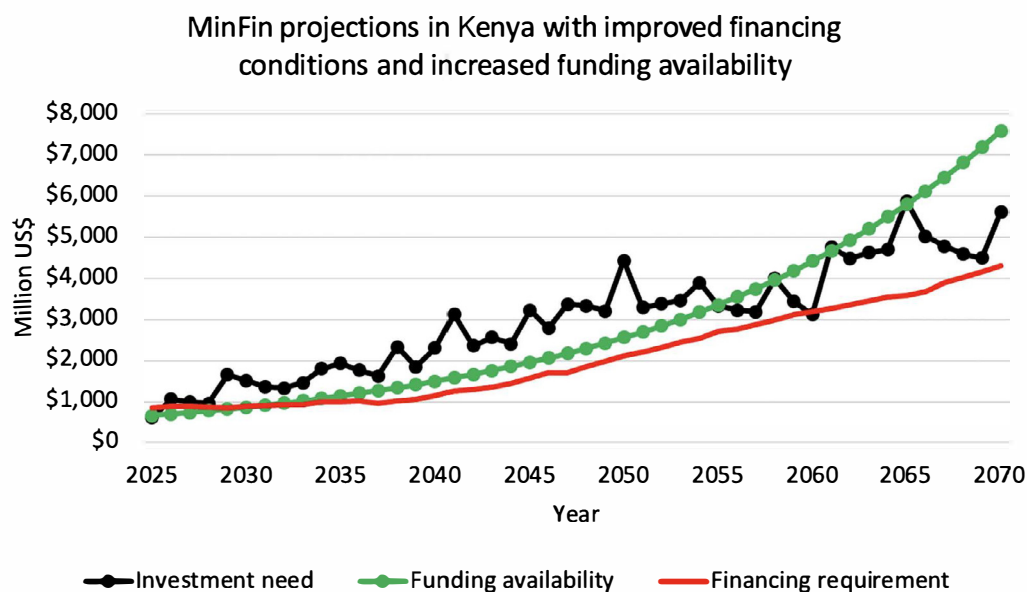


Figure 12: MinFin projections in Kenya with softer financing terms and increased funding availability. Note: The average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 30.0 years. The CAGRs for government spending, SOE, and grants are 5.5%, 5.5%, and 5.5%, respectively.



Morocco

Morocco has witnessed a notable 65 percent expansion in electricity generation over the past decade, driven by factors such as population growth, economic development, and initiatives aimed at improving electricity access. However, despite significant investments in innovative concentrated solar power (CSP) technology, the nation's energy landscape remains heavily reliant on fossil fuels, constituting almost 90 percent of the total primary energy supply and 80 percent of electricity supply (IEA, 2019). This dependence exposes the Kingdom to vulnerabilities such as price shocks and supply chain volatility (Preston, 2021).

In response to global environmental concerns, the Moroccan government has pledged to reduce greenhouse gas (GHG) emissions by 17 percent below business-as-usual (BAU) levels by 2030. The National Energy Strategy outlines an ambitious plan to expand renewable electricity to constitute 52 percent of installed capacity by the same year (IEA, 2019). However, Morocco faced challenges in meeting its 2020 renewable energy targets and struggled to achieve the Compound Annual Growth Rate (CAGR) necessary for renewables to meet the 2030 goal (Globaldata, 2022). Despite having abundant renewable resources, the Kingdom encounters difficulties in transitioning away from outdated, carbon-emitting technologies. Meanwhile, the country has continued to commission new coal plants (IEA, 2023c), emphasising the ongoing tension between environmental goals and energy needs. Beyond energy planning, uncertainties persist regarding Morocco's ability to attract sufficient capital flows, a critical factor in meeting projected energy demands and simultaneously achieving the ambitious energy transition goals set for its energy systems.

The OSeMOSYS model, adopted from Cannone et al., (2023), was employed to forecast investment requirements for Morocco. It should be noted that these results are based on starter data kits for OSeMOSYS, intended as initial modelling aids, and should not be construed as indicative of outcomes suitable for real-world policy decisions. These results are designed as a starting point for analysis and require further refinement and context-specific considerations for meaningful application in policy formulation.

OSeMOSYS findings from the NZ scenario reveal a significant upsurge of investment needs towards mid-century peaking at over US\$14 billion in 2050 – more than seven times the baseline level in 2025 (Figure 13). However, post-2050, following the surge necessary for electrifying Morocco's power sector, investment needs relatively rapidly revert to levels observed prior to the peak. Decarbonisation of power generation infrastructure entails substantial investment in onshore wind and solar technologies, alongside substantial grid enhancements to accommodate increased electrification. Despite the need to shift away from reliance on oil and gas-fired power generation plants, the compensation needed for the early shutdown of plants is relatively small.

Despite incurring higher initial capital costs due to increased investments in clean energy technologies, the NZ scenario generates substantial economic benefits primarily due to savings in carbon emissions (Figure 14). Economic benefits substantially outstrip incremental costs, except at the peak of the transition in the mid-2040s.

Figure 13: Technology-specific capital investment needs for achieving net zero in Morocco by 2050 as projected by OSeMOSYS.

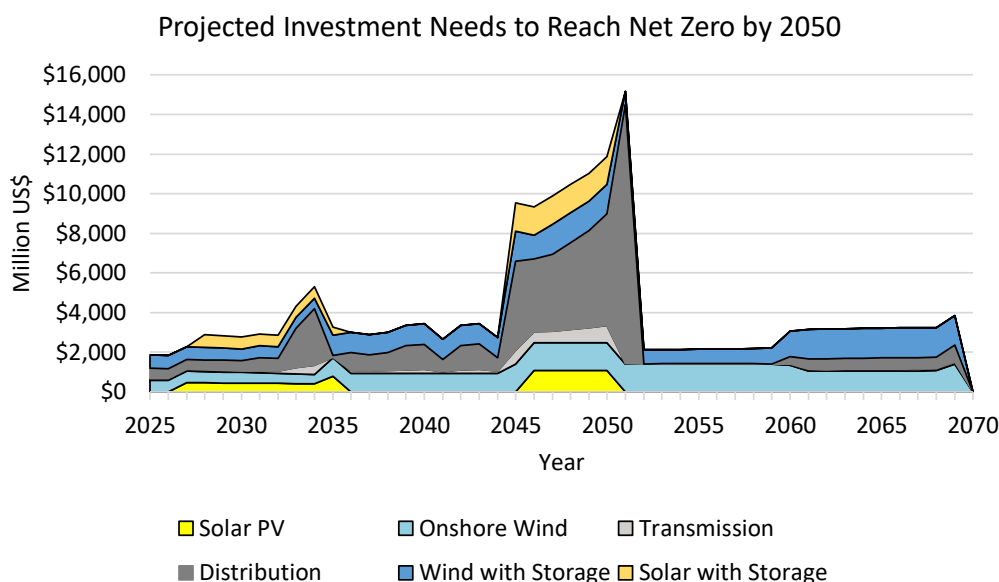
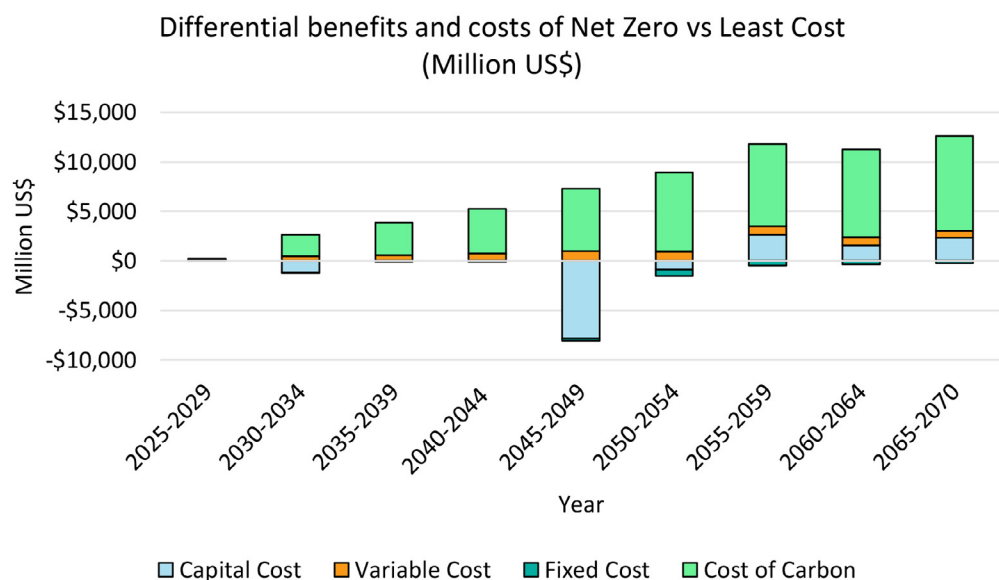


Figure 14: Differential benefits and costs of Net Zero vs Least Cost in Morocco, 5-year average.



The historic review of terms of financing terms for energy projects in Morocco over the period 2010–2023 finds that the country has on average attracted US\$986 million of finance for clean energy annually. Clean energy finance in Morocco is predominantly from international concessional sources (US\$639 million), with

softer financing terms, discounted at around 60 percent against market rates (Figure 15). Nevertheless, commercial finance – both from international and particularly domestic sources – has also made a substantial contribution (US\$127 and US\$220 million respectively), reflecting relatively favourable international investor

perceptions of country risk in Morocco, as well as greater maturity of domestic capital markets.

The weighted average financing terms for both debt and equity in Morocco's energy sector throughout the past decade featured an interest rate of 5.8 percent, coupled with a maturity period of 24.0 years and a grace period of 3.0

years. Once again, these terms vary widely across technologies. Overall, investments in transmission and distribution have faced a much higher cost of capital than those in generation (Figure 16), driven by a greater cost of equity used to finance domestic power utilities. Moreover, across all technologies, the cost of equity (15–35 percent) substantially outstrips that of debt (2–5 percent).

Figure 15: Merit order for historic finance sources of debt in Morocco, 2010 to 2023. Notes: Conc Dom = Domestic Public Sector concessional loans; Conc Intl = International Financial Institution concessional loans; Com Dom = Domestic commercial loans; Com Intl = International commercial loans.

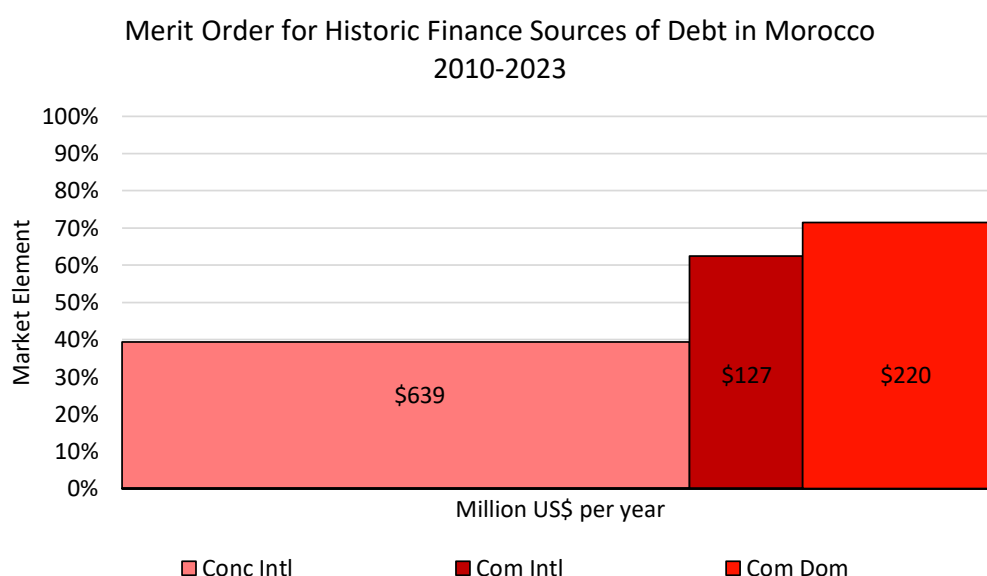
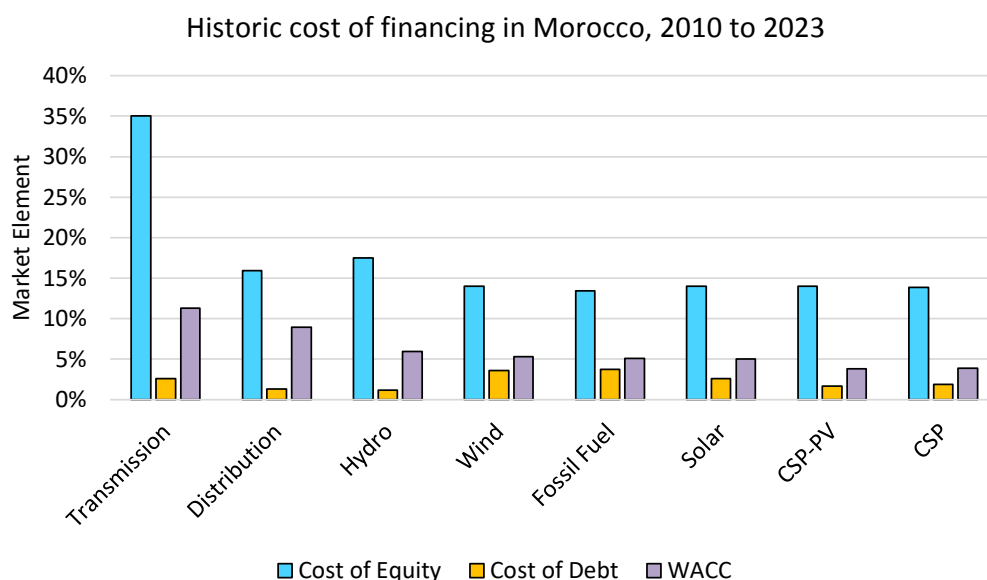


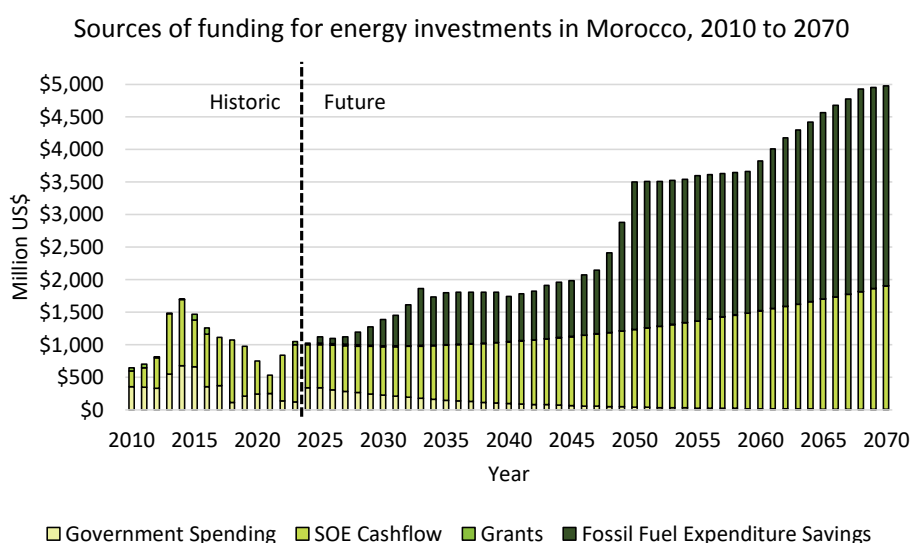
Figure 16: Historic Cost of Financing in Morocco based upon the period from 2010 to 2023.



Analysis of historic cashflows in Morocco highlights a notable decrease in public budget allocations coupled with a consistent strengthening of cashflow from energy utilities; with international cash grants playing a negligible role (see Figure 17). However, cashflows have experienced a decline, dropping from a peak of approximately US\$1,600 million in 2014 to a trough of under US\$600 million by 2021. Although recovering slightly thereafter, they have overall failed to keep pace with GDP growth during the same period. Nevertheless, over the entire period from 2010–2023 cashflow from energy utilities has been growing by 2.4 percent annually.

Looking ahead, the sustained importance of SOE generation as a funding source is projected (Figure 17). Concurrently, a major surge in funding availability is anticipated due to savings in fossil fuel purchases as Morocco phases out fossil fuel plants. These savings become particularly important from mid-century onwards, once the technology switch is complete, generating US\$2–3 billion of savings annually, which substantially exceeds historic cashflow trends. This makes it particularly important to ensure that these substantial savings are captured and redirected towards repayment of financing for renewable energy infrastructure.

Figure 17: Historic sources of funding availability for energy investments in Morocco. Notes: SOE = state-owned enterprise.



Bringing together these elements into the MinFin dashboard for Morocco, once again highlights that a net zero transition cannot be financed under business-as-usual financing and funding conditions. From 2035 onwards, a major gap opens up between financing needs and funding flow. This gap grows continuously, narrowing slightly around 2050 as savings in fossil fuel expenditure boost the availability of funding, but widening again to reach US\$6 billion per year thereafter before gradually levelling off (Figure 18). These results highlight the importance of

improving upon historic financing conditions, and/or boosting sector cashflows if the net zero transition is to be successfully implemented.

A series of simulations illustrate how this might be achieved. The first simulation considers a substantial softening of financing terms towards a weighted average cost of capital of 3.0 percent with an average financing term lengthening to 30 years with as much as 5.5 years of grace. Such a major reduction in the cost of capital is able to sustain financial equilibrium until mid-century,

but thereafter another substantial financing gap opens up (Figure 19). The second simulation leaves financing terms at their historic level but considers instead a steeper growth in sector cashflows that aligns with projected GDP growth of 6.5 percent, which allows Morocco to remain close to financial equilibrium throughout the period (Figure 20). A final simulation considers a combination of some improvement in financing terms including a 3.0

percent interest rate, an average grace period of 5.5 years, and an average finance term of 26.5 years. The simulation also assumes more conservative growth rates for cashflow, with Compound Annual Growth Rates (CAGRs) for government spending, State-Owned Enterprises (SOE), and grants set at 5.8 percent each. Results suggest that this combination of measures would also be successful in sustaining financial equilibrium (Figure 21).

Figure 18: MinFin projections in Morocco in absolute terms. Note: The weighted average interest rate is 5.8%, the average grace period is 3.0 years, and the average term of finance is 24.0 years. The CAGRs for government spending, SOE, and grants are -2.8%, 2.4%, and -9.5%, respectively.

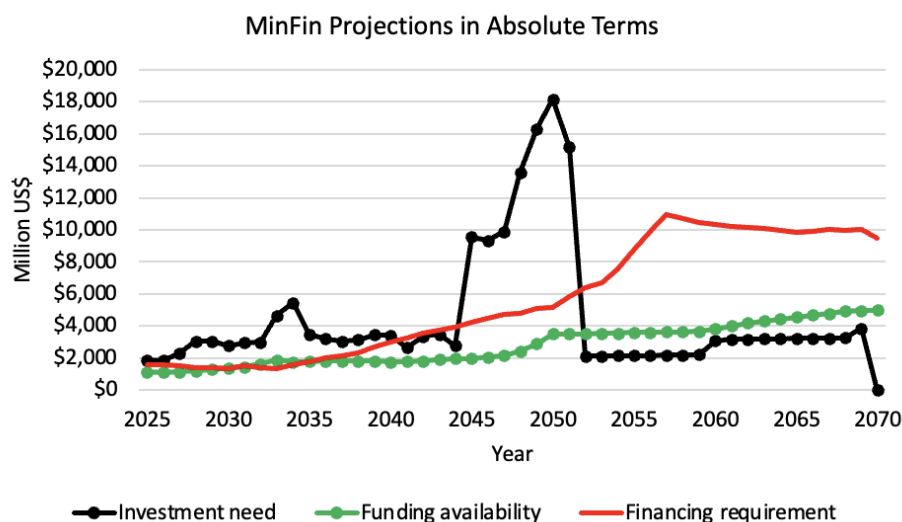


Figure 19: MinFin projections in Morocco with improved financing conditions. Note: The weighted average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 30.0 years. The CAGRs for government spending, SOE, and grants are -2.8%, 2.4%, and -9.5%, respectively.

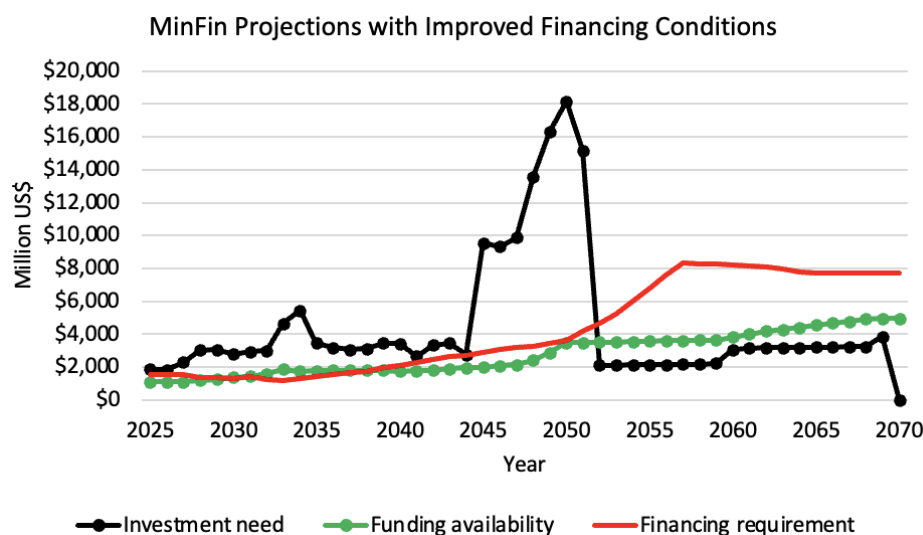


Figure 20: MinFin projections in Morocco with increased funding availability. Note: The weighted average interest rate is 5.8%, the average grace period is 3.0 years, and the average term of finance is 24.0 years. The CAGRs for government spending, SOE, and grants are 6.5%, 6.5%, and 6.5%, respectively.

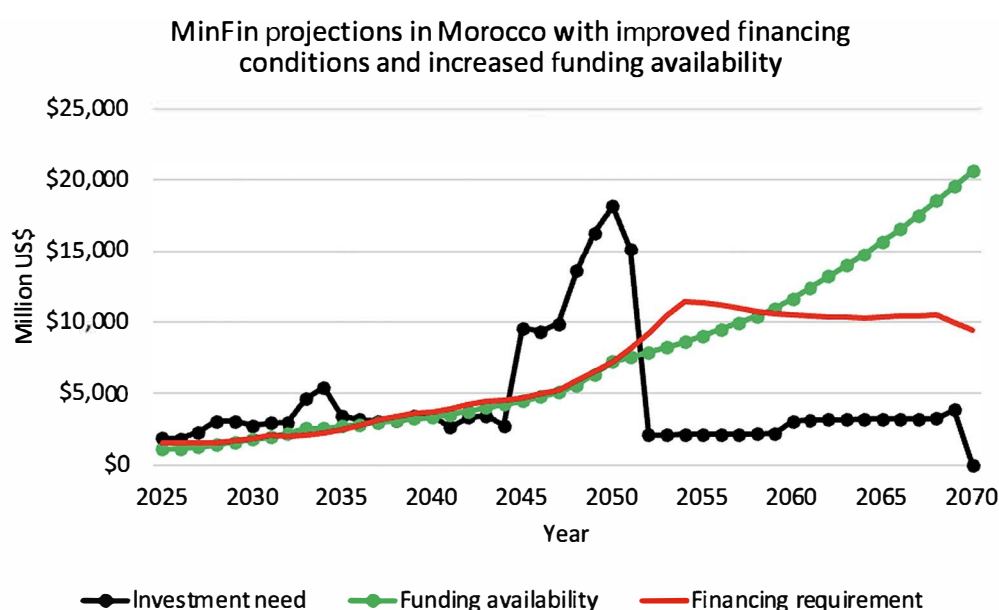
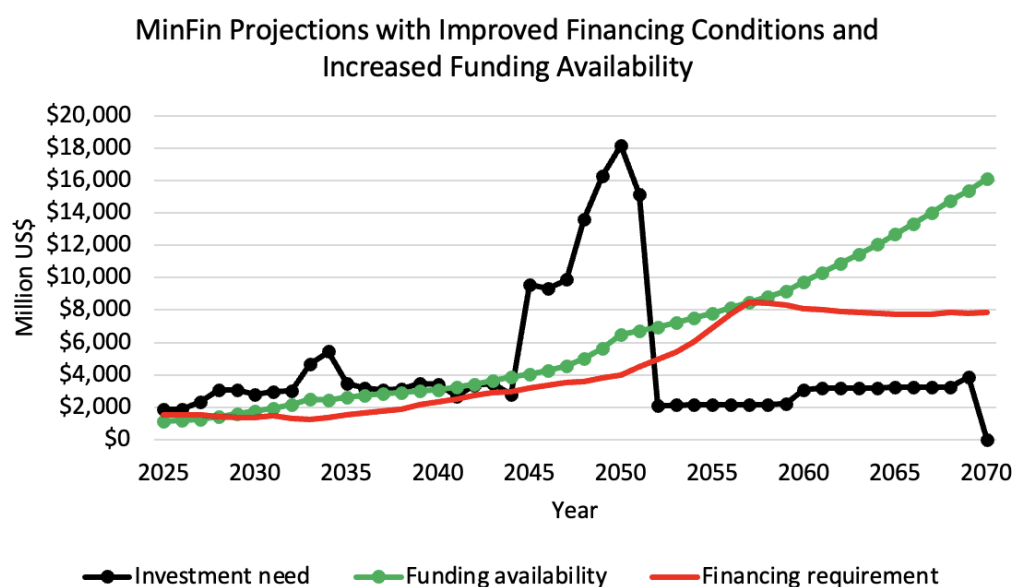


Figure 21: MinFin projections in Morocco with softer financing terms and increased funding availability. Note: The weighted average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 26.5 years. The CAGRs for government spending, SOE, and grants are 5.8%, 5.8%, and 5.8%, respectively.



Comparison

A comparison between the two cases brings several interesting matters to light.

First, normalising for the different sizes of the two economies, the net zero transition presents a more significant challenge for Morocco than for Kenya. This reflects the fact that Kenya's power system is already much more heavily skewed towards

renewable energy (90 percent versus 38 percent for Morocco) facilitating the energy transition. As a result, the cost of the net zero pathway does not exceed 0.6 percent of GDP throughout the transition process (Figure 22), while in Morocco the need for a more pervasive adjustment of the power generation portfolio makes the transition more burdensome, peaking at 3.0 percent of GDP by mid-century (Figure 23).

Figure 22: MinFin projections in Kenya with softer financing terms and increased funding availability, relative to GDP. Note: The average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 30.0 years. The CAGRs for government spending, SOE, and grants are 5.5%, 5.5%, and 5.5%, respectively.

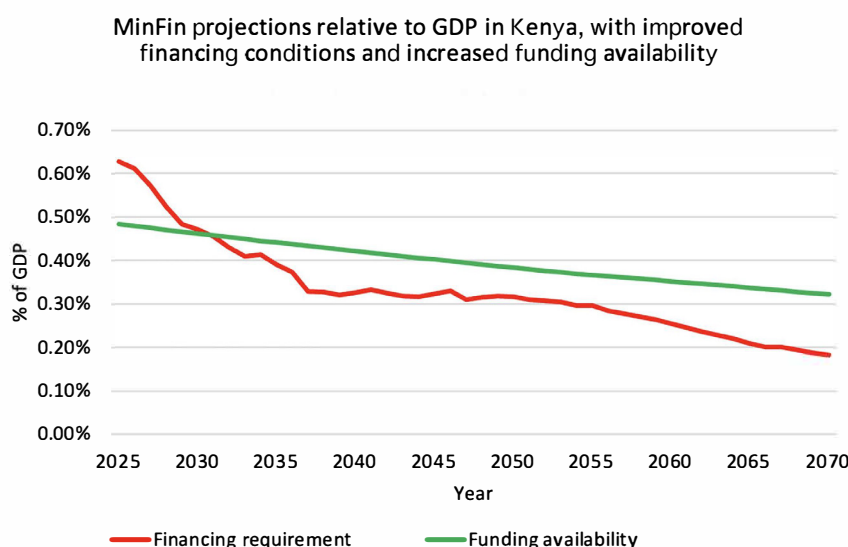
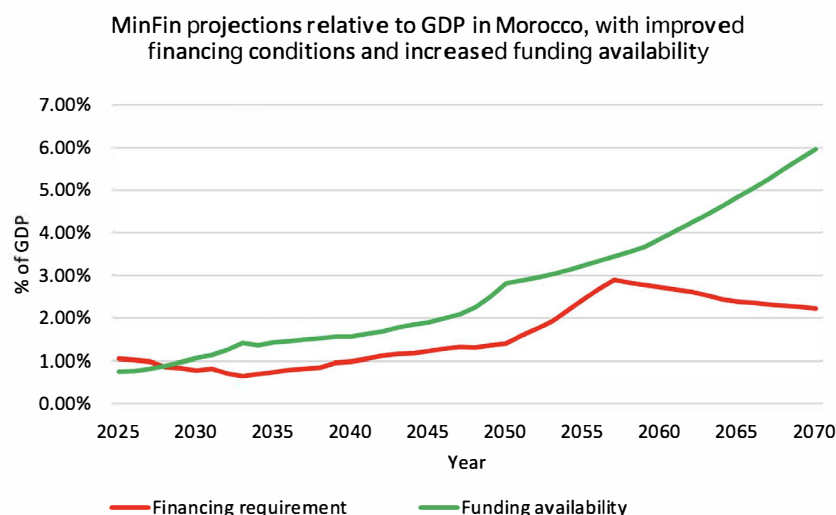


Figure 23: MinFin projections in Morocco with softer financing terms and increased funding availability, relative to GDP. Note: The weighted average interest rate is 3.0%, the average grace period is 5.5 years, and the average term of finance is 30.0 years. The CAGRs for government spending, SOE, and grants are 6.5%, 6.5%, and 6.5%, respectively.



Second, while both countries benefit from substantial volumes of international concessional finance, Kenya is far more dependent on such development banking sources which provide over 90 percent of financing (Figure 24). By contrast, Morocco obtains 35 percent of energy financing from commercial sources, most of that from the domestic capital market (Figure 25). This is reflected in the significantly higher share of private finance for energy projects of 40 percent in Morocco versus 15 percent in Kenya.

Third, despite the greater preponderance of concessional finance in Kenya, the overall WACC between the two countries is relatively similar. The weighted average interest rate for Kenya is 5.5 percent versus 5.8 percent for Morocco, and the weighted average maturity of loans is 25.6 years in Kenya versus 24.0 years in Morocco. An important reason for this is that Morocco is able to access commercial finance on significantly better terms than Kenya, with a weighted average interest rate for commercial finance of 4.3 percent versus 9.4 percent, reflecting a lower country risk rating of B versus BB+.

Figure 24: Historic sources of finance for energy investments in Kenya in 2010 to 2023.

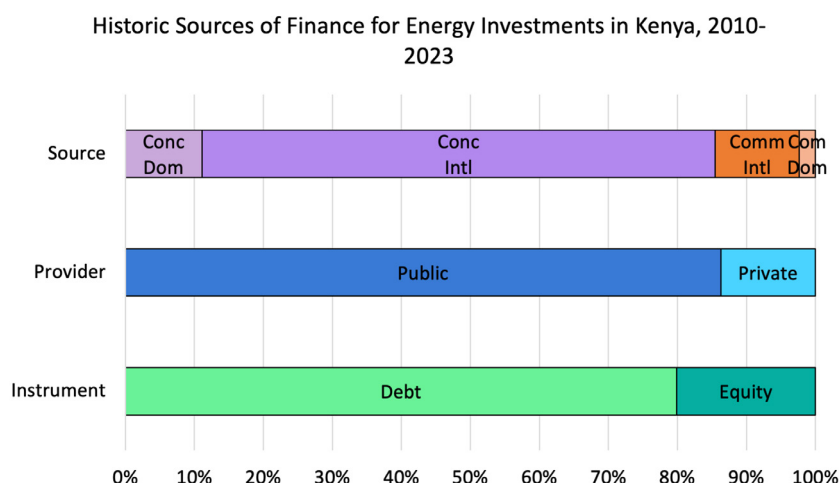
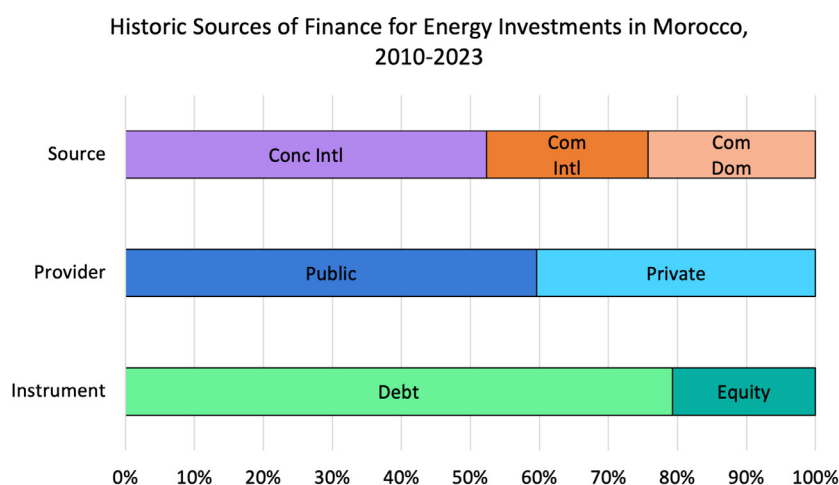


Figure 25: Historic sources of finance for energy investments in Morocco, 2010 to 2023.



Dom = Domestic Public Sector concessional loans; Conc Intl = International Financial Institution concessional loans; Com Dom = Domestic commercial loans; Com Intl = International commercial loans

6 CONCLUSIONS AND EXTENSIONS

In conclusion, nations embarking on the journey of transitioning from fossil fuels to a more sustainable energy landscape are poised to encounter a significant climate finance gap – a disparity between investment needs and available funding. This paper introduces a novel tool, MinFin, aimed at aiding Ministries of Finance in comprehending the potential climate finance shortfall, thereby enabling the formulation of effective finance strategies conducive to fostering climate-compatible growth.

These illustrative case studies demonstrate the feasibility of calibrating a MinFin model at the country level, and the value of the resulting insights. The cases consistently illustrate the severity of climate finance gaps faced by low- and middle-income countries under business-as-usual financing and funding conditions. In all cases, reaching net zero calls for a surge of investment needs towards mid-century that reaches several multiples of historic baseline levels. Countries differ significantly as regards their dependency on international concessional finance and their ability to access significant volumes of finance from domestic capital markets. With stagnant public investment in the energy sector, countries rely primarily on the cashflow of energy utilities to fund repayment of energy project finance. In addition, savings in fossil fuel expenditure were found to be a potential significant source of funding for fossil fuel dependent countries, such as Morocco.

Although preliminary, these simulations already indicate that the ability of LMICs to finance the climate transition is not something that can be taken for granted. Irrespective of global market conditions, the specific configuration of required

investment profiles, available financing terms and projected sector cashflows at the country level, critically determine financial feasibility. The two contrasting cases illustrate how rapidly this equation breaks down under the pressures generated by a mid-century net zero target. The implication is that efforts to improve available financing conditions, while boosting sector funding streams, are an essential part of the support that LMICs will need to deliver on climate objectives.

Going forward, this has several policy implications.

First, the financing terms and conditions available to LMICs have a significant impact on the affordability of implementing capital-intensive energy transition plans and need to be improved over time. This involves reducing country risk perceptions through an improved policy environment, as well as increasing the availability of concessional finance.

Second, another vital priority is to strengthen energy sector cashflows, particularly by improving the financial viability of the energy utilities that form the backbone of the sector and play a critical role as power purchasers for renewable energy projects. In practice, this means acknowledging the potential implications of net zero for energy pricing.

Third, efforts to support LMICs in planning for the energy transition should seriously examine the feasibility of countries successfully financing such plans, given the market conditions they face. The emerging MinFin model can provide a means for doing so, while facilitating collaboration between Ministries of Energy and

Finance, by translating sector investment plans into annual financing requirements.

Finally, the MinFin model is currently in the developmental stages and several extensions are under consideration. These include incorporating further analysis of economic and fiscal impacts, addressing the efficient allocation of finance, integrating more closely with other models, and broadening sector coverage. Although the primary application of these extensions will be an in-depth analysis for a limited set of countries, there is also the possibility for a broad and shallow application across a larger number of LMICs using the current MinFin model and a set of standardised open-source data kits.

Given the focus on a Ministry of Finance audience, the relevance of the MinFin model could be enhanced by the inclusion of further analytics covering economic and fiscal aspects of the transition. Accounting for economic benefits of climate investments helps to provide a more balanced case for the investments being made. It may also be of interest to incorporate employment multipliers from the different types of energy projects contemplated, and to examine the overall impact on energy costs and resulting distributional analysis using micro-simulation techniques. Fiscal impacts are also a very relevant consideration, given the extent to which fossil fuels are taxed and/or subsidised in many low- and middle-income countries, and changing fuel consumption patterns are available in the modelling framework. Debt sustainability is another concern that could potentially be incorporated into the model.

At present, MinFin focuses on whether financing and funding are in balance at the aggregate level. Another important question is how best to allocate available financing resources across the portfolio of required projects. Financing sources differ significantly according

to the concessionality of the resources that they provide. At the same time, investment projects differ significantly according to the concessionality of the resources that are needed to achieve financial viability. Incorporating a financial optimisation algorithm into MinFin would allow the model to provide guidance on the allocation of finance. Ideally, this would build on project-level financial analysis to determine the extent of viability gaps for different project categories.

MinFin is already an integrative model that draws upon inputs from OSeMOSYS and the FFRM. There is the potential to incorporate further model linkages. As noted above, connection to a project finance model would allow a deeper analysis of the allocation of available financing across projects. Furthermore, linkage of MinFin to a utility finance model would permit a more refined estimation of funding availability from utility cashflow, while linkage to a macroeconomic model would allow the impacts of the climate investment plan on economic growth to be taken directly into account.

While MinFin currently focuses on electricity, other sectors – such as heat, transport and industrial uses – are also relevant to the climate transition. These sectors are already covered under climate investment plans produced by OSeMOSYS, making it straightforward to incorporate the associated financing needs. However, where they may present significant challenges is in determining the financing and funding baselines. Project finance for these sectors tends to be smaller scale and dispersed, making it more challenging to document baseline financing conditions. At the same time, funding sources are similarly varied, with many coming from a large universe of households and corporations that cannot be readily documented.

In conclusion, while much attention to date has focused on the identification of optimal investment plans for the climate transition, a question of increasingly pressing concern is whether countries have the financial capacity to implement such plans, and what kind of support they may need from the international community. This paper presents an intuitive conceptual

framework for thinking about climate financing gaps at the country level and demonstrates how these can be empirically calibrated using a simple model. Emerging results indicate the potential significance of expanding this analysis to a larger number of countries, while at the same time incorporating further enhancements to the analytical framework.

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Appendix A: Projected Social Value of Carbon

The cost associated with carbon can be employed in assessing CO₂ emission outcomes obtained through OSeMOSYS. The presumed carbon cost is in accordance with the 2017 High-Level Commission on Carbon Prices, using midpoints between its high and low estimates.

Given that the High-Level Commission report does not explicitly outline the societal value of carbon beyond 2030, estimates from 2030 to 2050 are extrapolated using a consistently applied growth rate of 2.25% per year (Stiglitz et al., 2017).

Table 1: Cost of carbon applied to OSeMOSYS emissions results.

Year	Carbon Price (\$/Ton CO ₂ e)	Year	Carbon Price (\$/Ton CO ₂ e)
2025	67.06	2048	111.94
2026	68.57	2049	114.46
2027	70.11	2050	117.04
2028	71.69	2051	119.67
2029	73.30	2052	122.36
2030	75.00	2053	125.12
2031	76.69	2054	127.93
2032	78.41	2055	130.81
2033	80.18	2056	133.75
2034	81.98	2057	136.76
2035	83.83	2058	139.84
2036	85.71	2059	142.99
2037	87.64	2060	146.20
2038	89.61	2061	149.49
2039	91.63	2062	152.86
2040	93.69	2063	156.30
2041	95.80	2064	159.81
2042	97.95	2065	163.41
2043	100.16	2066	167.09
2044	102.41	2067	170.85
2045	104.72	2068	174.69
2046	107.07	2069	178.62
2047	109.48	2070	182.64

Appendix B: Data Collection Template for Financing Baseline

A comprehensive database that itemises each financier's share in each project and furnishes details regarding the financing terms has been

developed. These terms allow for the extraction of specific financial terms, such as PMT, annual repayment schedules, and concessionality.

Table 2: A list of data points required for the financing sources baseline in MinFin.

Name of Project	The name of the project receiving financing.
Name of Financier	The name of the financial organisation that is to carry out transactions.
Technology	The type of energy technology being financed, e.g., geothermal or biomass.
Year	The year in which the loan agreement is signed and finance is committed.
Financing Source	The classification of financiers by sources, encompassing (1) International Financial Institution concessional loans, (2) Domestic Public Sector concessional loans, (3) International commercial loans, and (4) Domestic commercial loans.
Type of Finance	The classification of the financing source as either debt or equity.
Financing Sector	The classification of the financing source according to whether public or private: public being finance provided by national and local government or development banks; and private finance being provided by individuals, businesses, or commercial investors.
Financial Institution	A more granular breakdown of financiers based on the following taxonomy: Bilateral Agency, Multilateral Agency, Foreign Government, National Government, National Development Bank, Climate Funds, Commercial Bank, and Private Equity Fund.
Origin of Finance	The geographic source of the financing, categories include International OECD, International China, International Other, and Domestic.
Volume of Finance	The principal value of the finance provided in Million US\$.
Rate of Return	The interest rate for loans, and for equity, the Expected Rate of Return (ERR).
Term	For a loan, the loan duration and; for equity, the project's lifespan.
Grace Period	The period from the issue year of a loan to the first repayment of principal.
Schedule	The schedule specifying the timing and amounts of interest and principal repayments required for a loan, which may be Equal Principal Payments (in which equal annual payments are made throughout the loan that incorporate both interest and principal repayment), Lump Sum Principal (in which interest is paid annually and principal repayment is made as a lump sum at the end of the loan), or Lump Sum Principal and Compound interest (where both principal and interest are fully repaid at the end of the loan) (Scott, 2017).
Start Year	The year in which the borrower is entitled to drawdown on the loan; loan year 0.

Appendix C: Standard Financial Formulas Used in MinFin

The derivation of specific financial terms, including PMT, annual repayment schedules,

and grant elements from the financial terms, is achieved as outlined below:

(1) PMT

$$PMT = \frac{pv * r}{[1 - (1 + r)^{-n}]}$$

Where:

r = Interest rate

pv = The present value; principle loan

n = Number of repayments per annum

(2) Grant Element

In the case of loans structured with lump sum repayment schedules, the formula used is expounded as follows:

$$\left(1 - \frac{\left(\frac{r}{n}\right)}{d}\right) * \left[1 - \left(\frac{\frac{1}{(1+d)^{n*g}} - \frac{1}{(1+d)^{(n*m)}}}{d * (n * m - n * g)}\right)\right]$$

Where:

r = Interest rate

m = Maturity

n = Number of repayments per annum

D = Discount Rate

p = Principle repayment periods (year – graceperiod)

g = Interval (year)

And:

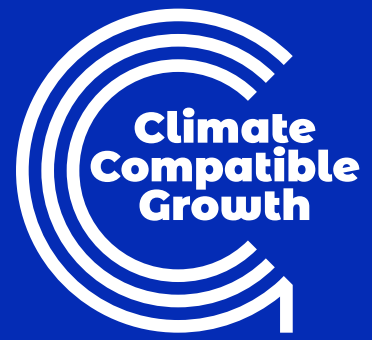
$$d = (1 + D)^{\frac{1}{n}} - 1$$

In the case of loans structured with lump sum repayment schedules, the formula used is expounded as follows:

$$1 - \frac{1 + r * m}{(1 + D)^m}$$

(3) Market Element

$$\text{Market Element} = 1 - \text{grant element}$$



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AUTHOR INFORMATION:

¹**Hannah Luscombe, Sam Fankhauser** (Smith School of Enterprise and the Environment, University of Oxford, Oxford, OX1 3QY)

²**Vivien Foster, Will Collier, Miles Greene** (Centre for Environmental Policy, Imperial College London, London, SW7 1NE, United Kingdom)

³**Jairo Quirós-Tortós** (School of Electrical Engineering, University of Costa Rica, San José, 11501-2060, Costa Rica)

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